Problem 3

Dr. Morty's adventure

40 minutes

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1 Scenario

You are a computational physicist. One evening, you go out for a walk with your laptop in your bag. You notice a dark alley, decide to explore it, and then suddenly everything goes blank. You wake up, and you really realise you are lying on the floor of a large observatory. You get up and notice two men in the discussion table with some instruments and large books. They seem to be arguing over some data. You walk up to them and introduce yourself as Dr Morty, a scientist. Within a few minutes of the conversation, you realise that the year is 1604, and the two men are Johannes Kepler and Tycho Brahe.

They are arguing over some model, suggested by Kepler, that doesn't fit the numbers. Then you enter their conversation, you tell them that you possess a kind of "computational engine", a tool capable of processing large calculations and testing hypotheses rapidly. You claim that, given the data alone, you can determine the correct laws and the structure governing the planetary motion. Tycho, who trusts the data more than the theory, is immediately interested. Kepler, although sceptical, is curious enough to agree.

They hand you the data, and now the problem is yours.

2 Tycho Brahe's problem

2.1 Motion of the planets and the Sun

The data set contains the Right Ascension, Declination, and distance from Earth (in AU) for each object at each observation over 3 Earth years. All distances are in astronomical units (AU). The Sun's distance from Earth is given (but you may note it is always about 1 AU). Right Ascension (RA) is in hours (0–24), Declination (Dec) in degrees (-90 to $+90$). The obliquity of the ecliptic is $\epsilon = 23.5°$. (**The data set is provided in github**)

1. The **Geocentric** model(a model of the solar system with the Earth at the centre)
   (a) Load the data and convert RA from hours to radians, Dec from degrees to radians. For each object, plot its RA vs Dec as a scatter plot. Use different colours for the Sun, Mars, Jupiter, and Saturn. Label the axes appropriately.
   (b) From the Data, first infer the time period of revolution of the different planets (in Earth Years)
   (c) Note that you are also given the distances of the Planets with respect to Earth. Using all this information, create a geocentric model of the Solar System.
2. The **Heliocentric** model (a model of the solar system with the Sun at the centre) of the Solar System
   (a) Given your geocentric model, now convert that into a heliocentric model of the solar system.
   (b) What can you infer about the Law of Gravity governing this universe?

3 Appendix: The equatorial coordinate system

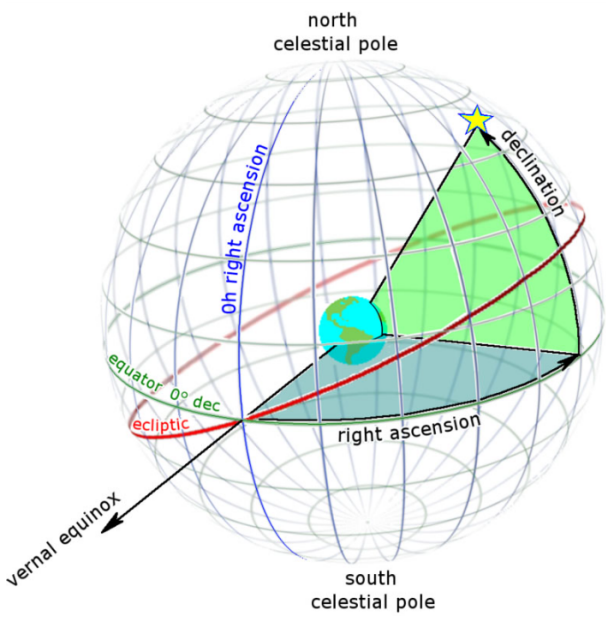

Figure 1: The equatorial coordinate system

1. Here, the Polaris (pole star) is located in the **north celestial pole** and at 90° from the Polaris is the celestial equator.
2. The **ecliptic** (or the great circle) is the motion of the sun, or from the sun would be the motion of the earth.
3. The angle between two stars (or any celestial objects) as seen from Earth is their angular separation on the celestial sphere. Given their equatorial coordinates (Right

Ascension and Declination), the angle θ between them is given by the spherical law of cosines:

$$\cos\theta = \sin\delta_1 \sin\delta_2 + \cos\delta_1 \cos\delta_2 \cos(\alpha_1 - \alpha_2)$$

where:

α_1, δ_1 = Right Ascension and Declination of the first star

α_2, δ_2 = Right Ascension and Declination of the second star

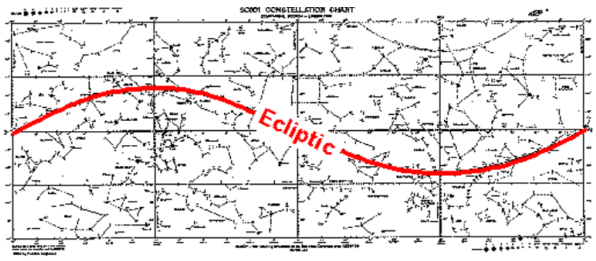

Figure 2: RA vs Dec map of Sun over 1 year (The ecliptic)

The transformation from equatorial coordinates to ecliptic longitude λ (assuming zero ecliptic latitude) is:

$$\sin\delta = \sin\epsilon \sin\lambda, \quad \cos\delta \cos\alpha = \cos\lambda, \quad \cos\delta \sin\alpha = \sin\lambda \cos\epsilon,$$

with $\epsilon = 23.5°$.